Squarepoint Online Assessment — Questions

Section 1 — Multiple Choice

Q1. TLB properties

Pick ONE or MORE

Which of the following are true about the TLB? Select all that apply.

- **(A)** It caches virtual-to-physical address translations
- **(B)** A context switch may require flushing it
- **(C)** A TLB miss can still be a page hit
- **(D)** It is managed explicitly by the application

Q2. Containers vs VMs

Pick ONE

A container and a virtual machine both run "Ubuntu 22.04" on a RHEL host. Which is true?

- **(A)** Both run their own Ubuntu kernel; the container's is just smaller
- **(B)** The container uses the host's RHEL kernel and supplies only the Ubuntu userspace; the VM runs its own Ubuntu kernel
- **(C)** The container requires the host itself to be running Ubuntu
- **(D)** The VM shares the host kernel and isolates only the filesystem

Q3. SQL - NULL handling

Pick ONE

Given `orders(id, customer_id, amount)` where some rows have a `NULL` value for `amount`, what does the following query do with the `NULL`s?

```
SELECT COUNT(*), COUNT(amount), SUM(amount)
FROM orders;
```

- **(A)** COUNT(*) counts all rows; COUNT(amount) and SUM(amount) skip NULLs
- **(B)** All three skip NULLs
- **(C)** SUM returns NULL if any row is NULL
- **(D)** All three count NULLs

Q4. Database Normalization

Pick ONE

`R(A, B, C, D)` has the functional dependencies `AB -> C` and `C -> D`. The candidate key is `AB`. What is the highest normal form that `R` satisfies?

- **(A)** 1NF
- **(B)** 2NF
- **(C)** 3NF
- **(D)** BCNF

Q5. C++ Virtual Functions

Pick ONE

What is printed on running the following code snippet?

```
struct Shape  { virtual std::string name() { return "Shape"; } };
struct Circle : Shape { std::string name() override { return "Circle"; } };

void byValue(Shape s)  { std::cout << s.name(); }
void byRef  (Shape& s) { std::cout << s.name(); }

int main() {
    Circle c;
    byValue(c);
    byRef(c);
}
```

- **(A)** ShapeCircle
- **(B)** CircleCircle
- **(C)** ShapeShape
- **(D)** CircleShape

Q6. Linux Process

Pick ONE

A process state shows as `D` in the output of `ps aux`. `kill -9 <pid>` has no effect. Why?

- **(A)** The process is a zombie; only the parent can reap it
- **(B)** It is in uninterruptible sleep on I/O; signals are only delivered when it returns to a schedulable state
- **(C)** SIGKILL was blocked by a signal handler
- **(D)** The process is running as root

Q7. Semaphores vs mutex

Pick ONE or MORE

Which are true of counting semaphores and mutexes?

- **(A)** A mutex has an owner — only the thread that locked it may unlock it
- **(B)** A counting semaphore can be signalled by a thread that never waited on it, which makes it suitable for producer-consumer signalling
- **(C)** sem_wait on a semaphore with value 0 blocks until another thread posts
- **(D)** A binary semaphore and a mutex are interchangeable in all cases

Section 2 — Coding

Q8. REST API: Energy Bar

Language: Python 3

Given a country name, find the chocolate from that country that provides the most energy per gram using a chocolate database.

Access the database with HTTP GET requests to `https://jsonmock.hackerrank.com/api/chocolates?countryOfOrigin={country}` (replace `{country}`). Results are paginated and accessible by appending `&page={num}` to the URL (replace `{num}`).

The API response includes these fields:

- *page*: current page
- *per_page*: maximum results per page
- *total*: total records

- *total_pages*: total pages for the query
- *data*: array of chocolate information

Each data object contains:

- *countryOfOrigin*: country of origin
- *brand*: brand name
- *type*: chocolate type
- *nutritionalInformation*: object with nutritional data
- *prices*: array of different prices
- *weights*: array of different weights
- other details not relevant to this question

Example record excerpt:

```
{
    "type": "Truffle",
    "brand": "Tony's Chocolonely",
    "ingredients": [...],
    "prices": [614, 283, 564, 365, 689],
    "weights": [125, 71, 116, 100, 126],
    "countryOfOrigin": "Norway",
    ...
}
```